Sungwon (Daniel) Lee

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EDUCATION

University of Michigan - Ann Arbor Ann Arbor, MI
Bachelor of Engineering - *Materials Science, minor in Electrical Engineering* **(Junior Standing)** *Aug 2025 - May 2029*
Cumulative GPA: 3.4/4.0;
Relevant courses this semester: Introduction to Materials, Logic Design, Programming & Intro Data Structures
Relevant courses next semester: Circuits, Computer Organization, Physics of Materials

WORK EXPERIENCE

Undergraduate Research Assistant - Selected Participant for TSMC Undergraduate Research Program Ann Arbor, MI
Li+ Research Group *May 2026 – Present*

- Fabricated next-generation ECRAM devices by executing **thin-film processing** (sputter deposition and furnace annealing) to integrate functions of both DRAM and NAND flash.
- Layered advanced **Tantalum, Yttria-Stabilized Zirconia (YSZ), and TaOx** films using cleanroom processing techniques to build high-performance memory stacking architectures
- Utilized **probing** machine to measure resistance inside memory device, aiming to reduce resistance and increase current to 60 mA
- Developing automated Python data-cleaning pipelines to parse raw experimental datasets, aiming to significantly decrease data processing turnaround time for the research team

Undergraduate Research Assistant Ann Arbor, MI
NOME Lab *August 2025-May 2026*

- Characterized crystallographic microstructures and grain boundary orientations across 20 ferric-martensitic (FM) steel alloys by operating **SEM and EBSD instrumentation**
- Logged **50+ hours** of hands-on operation on the Zeiss Crossbeam Focused Ion Beam (FIB)-SEM to gather high-resolution microstructural imagery

PROJECTS

Weather Balloon *Jan 2026 - April 2026*

- Developed & deployed an integrated sensor payload (barometric pressure, temperature, altitude) utilizing **Arduino/C++** for atmospheric data collection with a custom **6-layer PCB designed in KiCAD & soldered**
- Engineered power regulation and thermal casing to maintain electronics stability under near-space atmospheric conditions (10km in air)
- Parsed spatial log data post-flight with MATLAB to reconstruct vertical temperature profiles and flight trajectories

Subway Route Printed Circuit Board (PCB) Art *Aug 2026-Present*

- Designing **custom 2-layer PCB** art of transit maps using exposed ENIG gold copper traces for rail lines and surface-mount 0603 LEDs for key stations
- Integrating an ESP32-C3 microcontroller and USB-C power delivery to pull real-time API transit data over Wi-Fi, dynamically illuminating LED train locations
- Setting up & Designing Hong Kong MTR system, then will move onto New York

ACTIVITIES

U-M TAEKWONDO Ann Arbor, MI
4th Degree Black Belt *Aug 2025 - Present*

- Cooperate & coach ~40 students per session with sparring and kicking drills

SKILLS

Hardware & Cleanroom: Thin-Film Sputter Deposition, Wafer Probing, Furnace Annealing, SEM, EBSD, PCB Soldering, Multimeter, Oscilloscope
Programming & Software: C++, MATLAB, Python, Arduino, KiCAD, Fusion360, Microsoft Office
Languages: English, Korean